PSCo Hourly Demand Analysis

Project report | EIA Form 930 | 2019–2025 | Repository: https://github.com/ivky03/psco-grid-analytics

Project summary. This project examines hourly electricity demand reported for Public Service Company of Colorado (PSCo). A Python pipeline scanned 3,848,038 rows across 14 public EIA files (616,148,473 bytes), retained 61,368 PSCo hourly records, checked data quality, and stored the analysis table in SQLite. On a chronological test period, a gradient-boosting model had lower one-hour-ahead error than the two historical-demand baselines evaluated.

Purpose and scope

The goal was to create a reproducible analysis of PSCo demand data: validate the hourly series, summarize demand patterns, and test whether a simple machine-learning model improves on straightforward historical baselines. The data is reported at the balancing-authority level; it is not customer-level electricity consumption.

Data and processing

Item	Details
Source	U.S. Energy Information Administration Form EIA-930 public bulk CSVs
Period and files	2019–2025; 14 six-month files
Raw source scan	3,848,038 rows; 616,148,473 bytes
Selected analysis data	61,368 PSCo hourly records; timestamps in UTC
Ingestion method	pandas reads 50,000-row chunks and filters for PSCo

The pipeline converts timestamps and demand fields to usable types, keeps the reported raw demand value, and marks missing or nonpositive demand as unavailable for analysis. It writes the selected hourly table to SQLite, then runs SQL queries for data quality, peak demand, hourly and monthly profiles, and day-ahead forecast error.

Data quality

Check	Result	Handling
PSCo records	61,368	Seven years of hourly records
Duplicate timestamps	0	None detected
Missing hour slots	0	No gaps in the timestamp sequence
Missing demand values	24	Excluded from demand analysis
Nonpositive demand values	13	Excluded; raw readings retained

In total, 37 observations with missing or nonpositive demand were excluded from demand analysis. The timestamps remain present, which is why the series can have zero missing hour slots while some demand values are unavailable.

Forecast evaluation

The model predicts demand one hour ahead. Evaluation uses a chronological 80/20 split: earlier observations train the model, and the later 12,233 hourly observations form the held-out test period. This is a rolling evaluation, so the actual demand from the immediately preceding hour is available when predicting the next hour.

The gradient-boosting model uses hour, weekday, month, recent demand, demand at the same hour on the prior day, and the trailing 24-hour average. It is compared with previous-hour persistence (predict the next hour equals the latest observed hour) and a previous-day-same-hour baseline.

Method	MAE (MW)	RMSE (MW)	Horizon
Gradient boosting	103.72	146.03	One hour ahead
Previous-hour persistence	181.26	230.17	One hour ahead
Previous-day same hour	343.59	455.42	One-hour target
EIA reported forecast	383.39	481.10	Day ahead

Interpretation

On this held-out period, the model's mean absolute error was about 42.8% lower than previous-hour persistence and 69.8% lower than the previous-day baseline. These results are encouraging for this evaluation window. The EIA forecast is day-ahead, so its error is shown for context and is not a like-for-like comparison with the one-hour methods.

Scope and next step

The reported scores reflect one chronological holdout and a rolling one-hour-ahead setup using recent actual demand. Testing additional time periods and incorporating weather data would broaden the analysis.

Reproducibility

With Python dependencies installed, run the following from the project directory to download the public files and reproduce the multi-year pipeline:

```
python -m src.pipeline --mode real --start-year 2019 --end-year 2025
```

The run writes quality, ingestion-volume, and evaluation results to outputs/results.json. The EIA files are public; no API key is required.